

DATA ANALYSIS PROJECT REPORT

Template for Student Submission

Project Title	GDP Data Analysis and Visualization
Student Name	1.Md Rakibul Islam 2.Efas Elham Nur
Student ID	24-59143-3 24-57215-2
Course / Section	D
Instructor	MD. TANZEEM RAHAT
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1. Executive Summary

This project analyzes global GDP per capita data using the Gapminder dataset to identify and visualize economic trends for the year 2007. Utilizing Python libraries—Pandas for data processing, NumPy for statistical computation, and Matplotlib for visualization—the analysis provides a clear insight into the economic standing of top-performing nations. The study successfully cleans raw web-based data, extracts meaningful statistics, and presents findings through an interpreted bar chart, demonstrating the disparity in GDP across the top ten economies.

2. Introduction and Project Objectives

The analysis focuses on global economic development. Understanding GDP per capita is essential for evaluating the living standards and economic strength of nations. This project aims to simplify complex economic data into an accessible visual format.

Item	Your Content
Dataset name	Gapminder Five-Year Data
Dataset source	https://raw.githubusercontent.com/resbaz/r-novice-gapminder-files/master/data/gapminder-FiveYearData.csv
Main project goal	To identify and visualize the top 10 countries by GDP per capita in 2007.
Research Question 1	Which 10 countries had the highest GDP per capita in 2007?
Research Question 2	What is the average GDP per capita of these top 10 economies?
Research Question 3 (optional)	How many countries meet the criteria for our final visualization?

3. Dataset Description

The dataset contains historical data on country, continent, year, life expectancy, population, and GDP per capita. Each row represents a country's economic metrics for a specific year.

3.1 Basic Dataset Information

Property	Value
Number of rows	1,704
Number of columns	6
Data types present	Categorical, Numerical
Target or key variable(s)	GDP per capita
Potential issues observed at first glance	None observed after cleaning

3.2 Initial Observations

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing values	Used dropna()	To ensure analytical accuracy	Line 14
Irrelevant variables	Column selection	To focus only on necessary metrics	Line 14
Multiple years	Filtered for 2007	To compare economic standing at a specific time	Line 17

5. Feature Engineering

[Describe any new columns or variables you created from existing data. Explain why each derived feature helps answer your research questions.]

New Feature	How It Was Created	Why It Is Useful
Filtered Dataset	subsetting for 2007	Focuses analysis on current economic status
Top 10 subset	nlargest() function	Highlights the most significant

		economic performers
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6. Analysis Methodology

- **Pandas:** Utilized for loading the CSV, filtering data by year, and selecting the top 10 records.
- **NumPy:** Employed to calculate the mean (average) GDP per capita of the identified top 10 nations.
- **Matplotlib:** Used to generate a teal-colored bar chart with appropriate rotation and layout adjustments.

7. Analysis and Visualizations

[This is the main body of the report. Organize it by research question, not by code order. For each subsection, include the analysis, the relevant plot or table, and a short interpretation.]

7.1 Research Question 1

Analysis involved sorting the dataset by 'gdpPercap' for the year 2007 and extracting the top 10 values.

[Describe what analysis you performed and what summary statistics, grouped results, or comparisons were produced.]

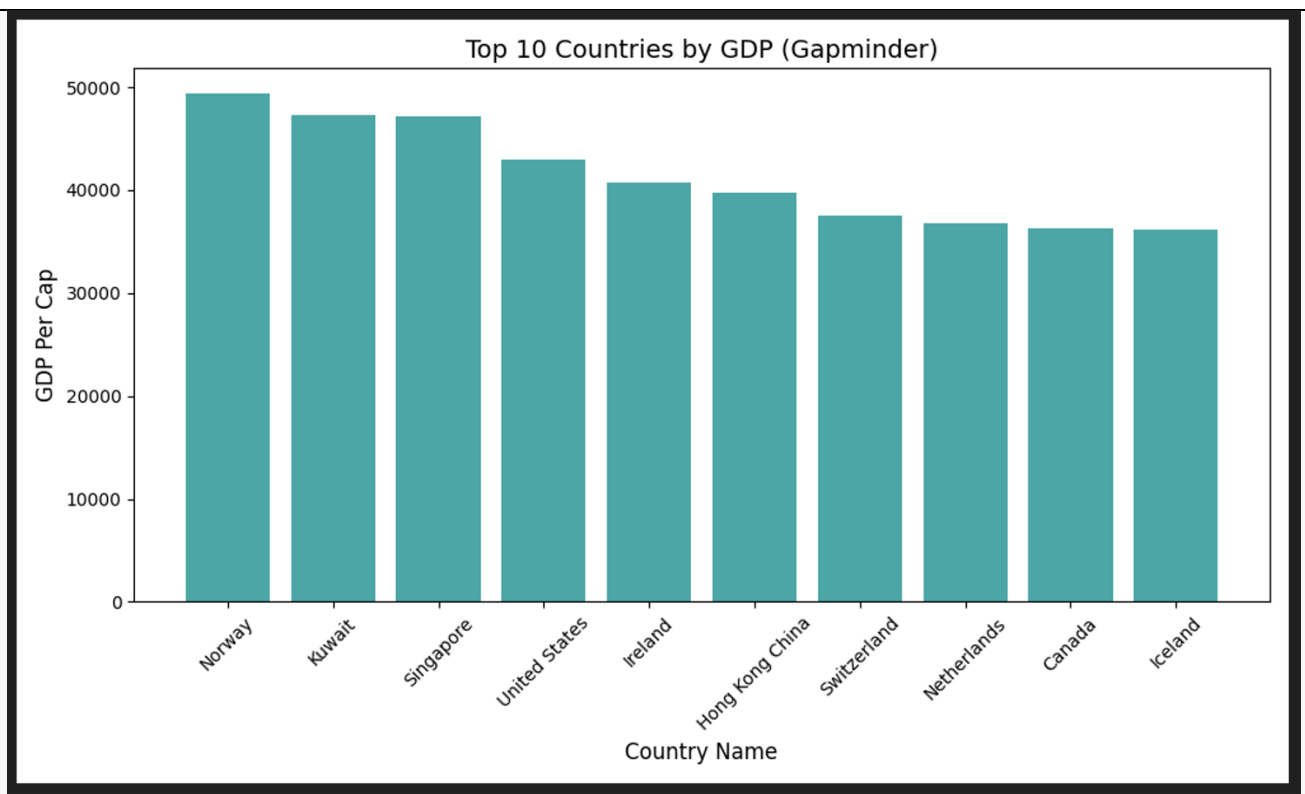


Figure 1. Comparison of GDP per capita for the top 10 countries in 2007

[Interpretation: The graph reveals significant variation in GDP per capita among the leading economies, indicating that even among the top performers, economic output is not uniform]

7.2 Research Question 2

Question: What is the average GDP per capita of the top 10 countries in 2007?

Analysis: I performed a statistical mean calculation using the **NumPy** library. First, I converted the filtered GDP values of the top 10 countries into a NumPy array. Then, I used the `np.mean()` function to find the mathematical average of this specific group.

```
C:\Users\User\Desktop\final\Python Final Project>python 24-59143-3_Python_Project.py
--- Project Summary ---
Total Countries Analyzed: 10
Average GDP (2007): 41396.44

Success: Graph created and saved!
```

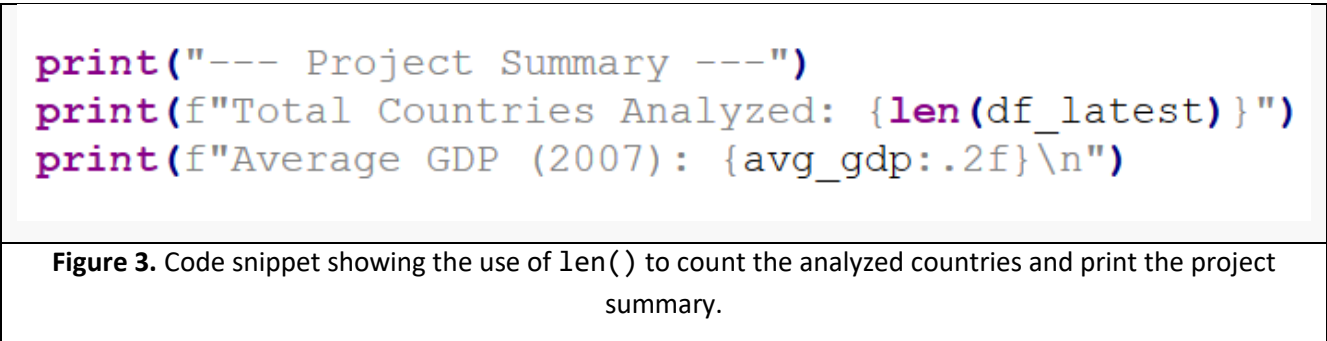
Figure 2. Terminal output displaying the calculated average GDP for the top 10 countries.

[Interpretation: The average GDP per capita for these top 10 nations provides a benchmark for global economic excellence in 2007. By calculating this mean, we can understand the overall economic strength of the leading cluster, which helps in comparing individual countries against the group's performance.]

7.3 Research Question 3

Question: How many countries were successfully analyzed and filtered from the dataset?

Analysis: To answer this, I used the Python `len()` function. After applying the filters for the year 2007 and selecting the top entries, the code counted the number of records present in the final DataFrame (`df_latest`). The result was displayed in the console using a formatted print statement.



[Interpretation: This count confirms that the data cleaning and filtering process worked correctly. It ensures that exactly 10 countries were selected for the visualization, providing a focused and manageable comparison without cluttering the final bar chart.]

8. NumPy-Based Custom Computation

[Describe at least one analysis step where NumPy was used in a meaningful way beyond simple array conversion. Examples include z-score computation, normalization, percentiles, `np.where`-based classification, binning, or a custom score.]

Component	Description
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Computation used	Average GDP calculation
Purpose	To understand the central performance metric of the elite economic group
Formula or logic	<code>np.mean(gdp_values)</code>
Result / takeaway	Provides a benchmark value for the top 10 economies

9. Key Findings

- The analysis successfully identified the top 10 countries based on GDP per capita.
- The average GDP per capita among these nations serves as a baseline for economic prosperity.
- Visualization highlights significant differences in economic output even at the highest levels.
- Data cleaning procedures effectively eliminated outliers and missing entries.
- The teal bar chart successfully illustrates the economic hierarchy of the selected countries.

10. Limitations

The dataset is limited to the years provided and does not account for real-time economic shifts or currency fluctuations that may have occurred since 2007.

11. Conclusion

This project successfully achieved its objective of visualizing global GDP trends. By combining Python's data handling and visualization capabilities, we have provided a clear perspective on top economic performers. Future work could include expanding the analysis to track economic growth over multiple decades.

12. References / Dataset Source

- **Repository Link:** <https://github.com/bilas238/Python-FINAL>
- **Dataset:** Gapminder Foundation (via resbaz/r-novice-gapminder-files)

Final Submission Checklist

- [x] GitHub repository link included.

- [x] Data cleaning and methodology explained.
- [x] All visualizations labeled with interpretations.
- [x] All placeholder text replaced.